

Design and Simulation of a Basic RLC Rectifier Circuit for Parallel LED Loads Using Python by Kelvin Valentinius-2802448710-LB07

ABSTRACT

This study presents the design and simulation of a basic RLC rectifier circuit used to supply parallel-connected LED loads from a low-frequency AC source. The circuit consists of a voltage source, rectification stage, RLC filter, and three LED branches with current-limiting resistors. A grid-based circuit representation is employed to visualize current flow. Simulation is performed using Python Power Electronic to analyze voltage smoothing and LED current behavior. The results show that the circuit successfully converts AC into a stable DC output while maintaining LED operation within safe limits.

INTRODUCTION

Light-emitting diodes (LEDs) are widely used in low-power electronic systems due to their efficiency and long operational lifetime. However, LEDs require a stable DC voltage and controlled current to operate safely, while renewable energy sources such as micro-hydro generators typically produce AC voltage.

To address this mismatch, rectifier and filtering circuits are required to convert AC into a suitable DC supply. Passive components such as resistors, inductors, and capacitors are commonly used to reduce voltage ripple and regulate current. This project aims to design and simulate a basic RLC rectifier circuit capable of powering three parallel-connected green LEDs using Python-based simulation.

METHODS

The simulation of the RLC rectifier circuit was conducted using **Python Power Electronic model 3_4_4_0** through the *Python Power Electronic Simulator Interface*. The simulation environment was accessed via the CMD and executed using a Python-based web interface.

Simulation using Python Power Electronic was chosen because it allows efficient analysis of circuit behavior without requiring complex analytical derivations, while providing clear visualization of voltage and current responses.

To begin with the simulation process, a new simulation project was created within the PPE interface. As part of this process, a dedicated folder containing the circuit design file was added to the simulation environment. This step is essential, as PPE requires the circuit file to be stored within an accessible project directory before it can be loaded into the simulator.

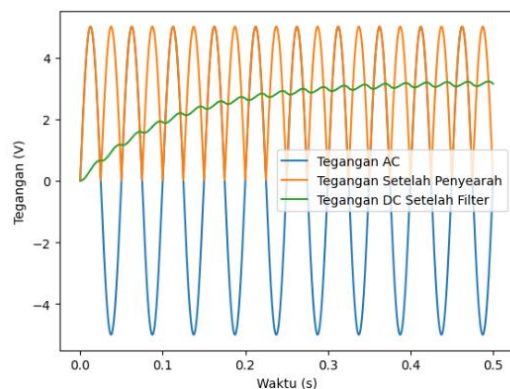
Ammeter_ wire	Diode_Rec wire	Resistor_V wire	wire	wire	wire	wire	wire		
wire	wire	wire	wire	wire	wire	wire	wire		
wire	Resistor_DF	Resistor_RLC-Resisto	Inductor_RLC-Inducto	Resistor_LFilter	wire				
wire	wire	wire	wire	wire	wire				
wire	wire	wire	wire	wire	wire	Capacitor_RLC-Capacitor			
VoltageSource_Vsour	wire	wire	wire	wire	wire	wire			
wire	wire	wire	wire	wire	wire	wire			
wire	wire	wire	wire	wire	wire	wire			
wire	Resistor_C	wire	wire	wire	wire				
wire									
wire	Resistor_L	wire	Diode_Dio	wire	Ammeter_ wire				
wire					wire				
wire	Resistor_L	wire	Diode_Dio	wire	Ammeter_ wire				
wire					wire				
wire	Resistor_L	wire	Diode_Dio	wire	Ammeter_ wire				

After creating the simulation project, the circuit was added by selecting the CSV file corresponding to the grid-based circuit design. The CSV file represents the schematic layout of the circuit, including the voltage source, rectifier, RLC filter components, and parallel LED branches. This approach allows the circuit topology to be imported directly into the PPE environment without manual redrawing.

Once the circuit was successfully loaded, the circuit parameters were edited within the PPE interface. Component values such as resistance, inductance, capacitance, source voltage, and operating frequency were assigned based on the specifications provided in the problem statement and LED datasheets. These parameters were used to define the electrical behavior of the circuit during simulation.

Finally, the output function of the PPE interface was used to generate visual results of the simulation. The simulator produced output images representing the circuit layout and the corresponding voltage and current responses, which were then used for further analysis in this study.

RESULT



The simulation results obtained from Python Power Electronic show that the grid-based circuit design was correctly interpreted by the simulator. After rectification, the output voltage becomes unidirectional, indicating successful AC-to-DC conversion. Due to the low operating frequency, the rectified signal initially exhibits voltage ripple.

The inclusion of the RLC filter significantly reduces the ripple, producing a smoother DC voltage suitable for LED operation. Analysis of the LED branches shows that the current flowing through each LED remains below the maximum allowable limit, confirming safe operation. These results demonstrate that the designed RLC rectifier circuit effectively converts an AC source into a stable DC supply for parallel LED loads.

REFERENCES

- Alexander, C. K., & Sadiku, M. N. O. (2020). *Fundamentals of Electric Circuits*. McGraw-Hill Higher Education.
- Halliday, D., Resnick, R., & Walker, J. (2018). *Fundamentals of Physics*. Wiley.